

We know that if we project (transform) the data of a class using a projection vector $\mathbf{u} \in \mathbb{R}^p$ to a p dimensional subspace ($p \leq d$), i.e.:

$$\mathbf{x} \mapsto \mathbf{u}^\top \mathbf{x}, \quad (46)$$

for all data instances of the class, the mean and the covariance matrix of the class are transformed as:

$$\boldsymbol{\mu} \mapsto \mathbf{u}^\top \boldsymbol{\mu}, \quad (47)$$

$$\boldsymbol{\Sigma} \mapsto \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u}, \quad (48)$$

because of characteristics of mean and variance.

The Fisher criterion (Xu & Lu, 2006) is the ratio of the between-class variance, σ_b^2 , and within-class variance, σ_w^2 :

$$f := \frac{\sigma_b^2}{\sigma_w^2} = \frac{(\mathbf{u}^\top \boldsymbol{\mu}_2 - \mathbf{u}^\top \boldsymbol{\mu}_1)^2}{\mathbf{u}^\top \boldsymbol{\Sigma}_2 \mathbf{u} + \mathbf{u}^\top \boldsymbol{\Sigma}_1 \mathbf{u}} = \frac{(\mathbf{u}^\top (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2}{\mathbf{u}^\top (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u}}. \quad (49)$$

The FDA maximizes the Fisher criterion:

$$\underset{\mathbf{u}}{\text{maximize}} \quad \frac{(\mathbf{u}^\top (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2}{\mathbf{u}^\top (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u}}, \quad (50)$$

which can be restated as:

$$\begin{aligned} &\underset{\mathbf{u}}{\text{maximize}} \quad (\mathbf{u}^\top (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2, \\ &\text{subject to} \quad \mathbf{u}^\top (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u} = 1, \end{aligned} \quad (51)$$

according to Rayleigh-Ritz quotient method (Croot, 2005). The Lagrangian (Boyd & Vandenberghe, 2004) is:

$$\mathcal{L} = (\mathbf{u}^\top (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2 - \lambda (\mathbf{u}^\top (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u} - 1),$$

where λ is the Lagrange multiplier. Equating the derivative of $\mathcal{L}$ to zero gives:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \mathbf{u}} &= 2(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2 \mathbf{u} - 2\lambda (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u} \stackrel{\text{set}}{=} \mathbf{0} \\ \implies (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2 \mathbf{u} &= \lambda (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u}, \end{aligned}$$

which is a generalized eigenvalue problem $((\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2, (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1))$ according to (Ghojogh et al., 2019b). The projection vector is the eigenvector of $(\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1)^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2$; therefore, we can say:

$$\mathbf{u} \propto (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1)^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2.$$

In LDA, the equality of covariance matrices is assumed. Thus, according to Eq. (18), we can say:

$$\mathbf{u} \propto (2\boldsymbol{\Sigma})^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2 \propto \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2. \quad (52)$$

According to Eq. (46), we have:

$$\mathbf{u}^\top \mathbf{x} \propto (\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2)^\top \mathbf{x}. \quad (53)$$

Comparing Eq. (53) with Eq. (23) shows that LDA and FDA are equivalent up to a scaling factor $(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ (note that this term is multiplied as an exponential factor before taking logarithm to obtain Eq. (23), so this term a scaling factor). Hence, we can say:

$$\text{LDA} \equiv \text{FDA}. \quad (54)$$

In other words, FDA projects into a subspace. On the other hand, according to Section 7, LDA can be seen as a metric learning with a subspace where the Euclidean distance is used after projecting onto that subspace. *The two subspaces of FDA and LDA are the same subspace.* It should be noted that in manifold (subspace) learning, the scale does not matter because all the distances scale similarly.

Note that LDA assumes *one* (and not several) Gaussian for every class and so does the FDA. That is why FDA faces problem for multi-modal data (Sugiyama, 2007).

9. Relation to Logistic Regression

According to Eqs. (16) and (32), Gaussian and Bernoulli distributions are used for likelihood (class conditional) and prior, respectively, in LDA and QDA. Thus, we are making assumptions for the likelihood and prior, although we finally work with posterior in LDA and QDA according to Eq. (15). *Logistic regression* (Kleinbaum et al., 2002) says why do we make assumptions on the likelihood and prior when we want to work on posterior finally. Let us make assumption directly for the posterior.

In logistic regression, first a linear function is applied to the data to have $\beta^\top \mathbf{x}'$ where $\mathbb{R}^{d+1} \ni \mathbf{x}' = [\mathbf{x}^\top, 1]^\top$ and $\beta \in \mathbb{R}^{d+1}$ include the intercept. Then, logistic function is used in order to have a value in range $(0, 1)$ to simulate probability. Therefore, in logistic regression, the posterior is assumed to be:

$$\begin{aligned} \mathbb{P}(\mathcal{C}(\mathbf{x}) | X = \mathbf{x}) \\ = \left(\frac{\exp(\beta^\top \mathbf{x}')}{1 + \exp(\beta^\top \mathbf{x}')} \right)^{\mathcal{C}(\mathbf{x})} \left(\frac{1}{1 + \exp(\beta^\top \mathbf{x}')} \right)^{1 - \mathcal{C}(\mathbf{x})}, \end{aligned} \quad (55)$$

where $\mathcal{C}(\mathbf{x}) \in \{-1, +1\}$ for the two classes. Logistic regression considers the coefficient β as the parameter to be optimized and uses Newton's method (Boyd & Vandenberghe, 2004) for the optimization. Therefore, in summary, logistic regression makes assumption on the posterior while LDA and QDA make assumption on likelihood and prior.

10. Relation to Bayes Optimal Classifier and Gaussian Naive Bayes

The Bayes classifier maximizes the posteriors of the classes (Murphy, 2012):

$$\hat{\mathcal{C}}(\mathbf{x}) = \arg \max_k \mathbb{P}(\mathbf{x} \in \mathcal{C}_k | X = \mathbf{x}). \quad (56)$$